My Reproducible Manuscript

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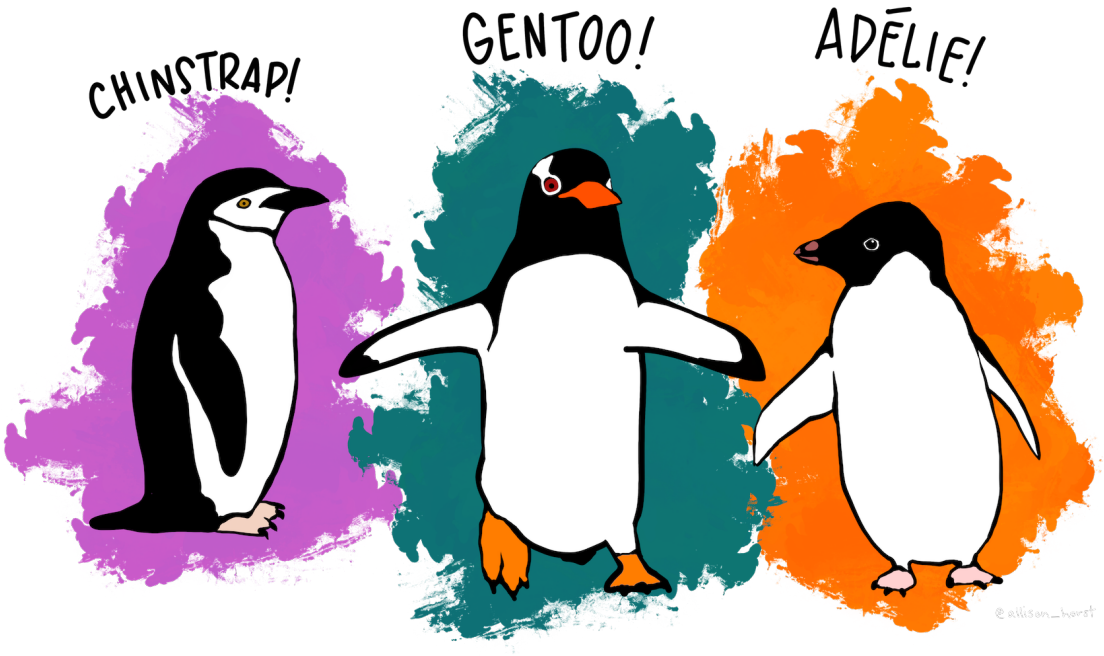
```
# Load penguins
library(ggplot2)
```

Meet Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see https://quarto.org.

Meet the penguins

The `penguins` data from the palmerpenguins package contains size measurements for 344 penguins from three species observed on three islands in the Palmer Archipelago, Antarctica.



Here's a citation: Horst et al. (2022)

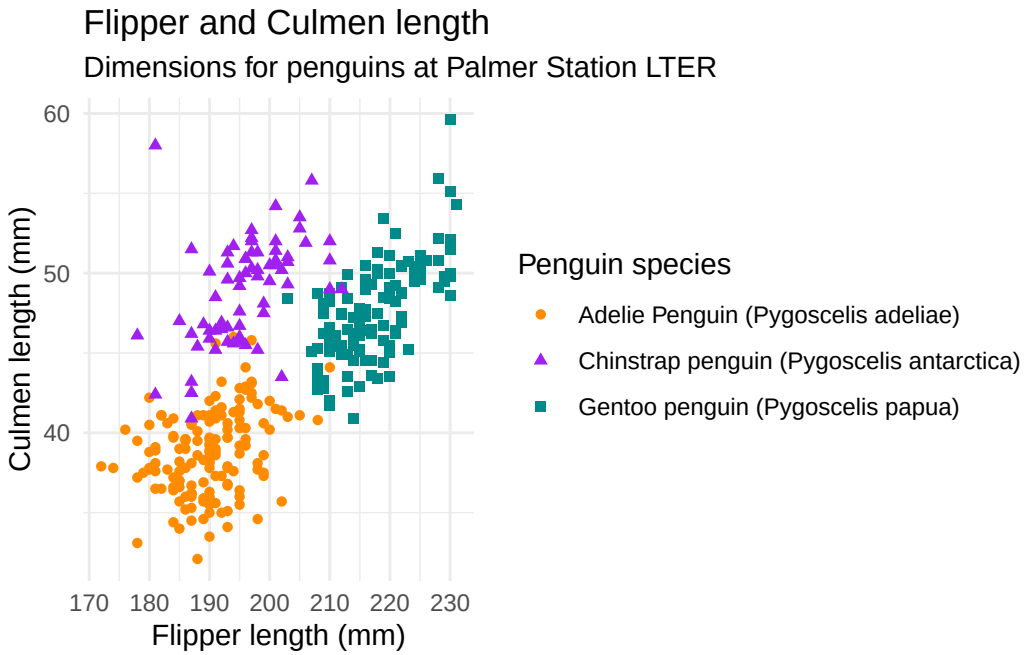
Results

```
# Read data
penguins <- read.csv("../data/data.csv")
## For some reason the original without "../" at the start didn't work, even
## though I've checked again and again that the working directory should be
## correct. It seems like it still wants to look for the file within the docs
## folder instead of the working directory and the prefix seems to fix that.
## I think this tells it to back out of the docs folder to the project folder,
## so that it can then look for the data folder there.
```

```
# Plot penguins
ggplot(penguins,
       aes(x = `Flipper.Length..mm.`, y = `Culmen.Length..mm.`)) +
        geom_point(aes(color = Species, shape = Species)) +
        scale_color_manual(values = c("darkorange","purple","cyan4")) +
        labs(
                title = "Flipper and Culmen length",
```

```
                subtitle = "Dimensions for penguins at Palmer Station LTER",
                x = "Flipper length (mm)", y = "Culmen length (mm)",
                color = "Penguin species", shape = "Penguin species"
        ) +
        theme_minimal()
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Doing a simple calculation within my text to calculate 50.

There are 124 Gentoo penguins, and 68 Chinstrap penguins in the data.

I don't think I need another in-line computation or calculation, but I guess we can calculate that 100 percent of the penguins in the data are in the Anvers region.

```
do_addition <- function(number1, number2){

        # Check that arguments provided are numberic
        if(class(number1) != "numeric" | class(number2) != "numeric"){
                print("Error: one or more of your inputs are not numeric")

        # Perform simple addition and return the result
        } else{
```

```
                result <- number1 + number2
                return(result)
        }
}
```

Now I can use this function to once again calculate that 25 + 25 = 50

I guess I can use this to define functions (e.g. for my simulation) in a different file than where I produce the rest of my thesis, so that I don't need to scroll over all of the simulation code when I'm just writing my thesis? Then in practice I would probably want to do something different with the combination of what is executed vs shown in the rendered file.

Horst, Allison M., Alison Presmanes Hill, and Kristen B. Gorman. 2022. "The R Journal: Palmer Archipelago Penguins Data in the Palmerpenguins R Package - An Alternative to Anderson's Irises." *The R Journal* 14 (1): 244–54. https://doi.org/10.32614/RJ-2022-020.